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Date issued: 02/09/2025

CS924: Designing a Strategic Agent for an Auction

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***Abstract*—This paper presents the design and implementation of a strategic bidding agent for a multi-round art auction. The primary objective of the agent is to acquire a complete collection of paintings while efficiently managing a constrained budget. To achieve this, the agent uses a Bayesian inference model to predict opponent bidding behaviours and compute the minimum competitive bid. A key feature of this model is an adaptive standard deviation algorithm that continuously updates the normal distribution based on a rolling window of recent winning bids, allowing the agent to dynamically adapt to varying adversarial strategies and aggressive bidding tactics. Furthermore, the paper discusses the exploration and subsequent rejection of a reinforcement learning approach, highlighting the specific challenges of state-space sparsity in environments with multiplicative scoring conditions. Finally, testing of the agent against a diverse set of adversaries, including random agents, clones, and purpose-built agents, helped to demonstrate the resilience and adaptability of the proposed agent in changing competitive environments. Firstly, a benchmarking test against a random bidder was done, which yielded a 100 % win-rate. This shows the agent's improvement over random bids. Then, of seven further tests, the agent obtained 100 % win-rate in three of them, and 70 % in one, although results degraded in tests with larger numbers of purpose-built adversaries.**

***Index Terms*—agents, machine learning, q-learning, game theory, inference**

I. Introduction

Multi-round, multi-agent auctions present complex and compelling challenges from a game-theoretic perspective. In an auction environment where agents compete to collect works by four distinct artists, the core challenge is determining the precise moment and amount to bid. Bidders must constantly weigh the trade-off between bidding aggressively to secure a specific painting and bidding conservatively to preserve their budget for future rounds. To navigate this environment, strategic agents can employ many principles and methods from statistics, machine learning, decision theory, and game theory to formulate an optimal strategy and execute the best response to dynamically changing auction conditions.

This report discusses the theoretical foundation behind the implemented bidding pipeline for a strategic autonomous agent participating in this multi-round, first-price, sealed auction. In **Section II**, the various theoretical aspects and methodologies of the agent are detailed, illustrating how these components interact to compute a highly strategic final bid.

II. Implementation

The implemented strategy used has three main components. They will be outlined in order of computation, as a trace through the algorithm. First, the Bayesian inference computes a bid using Bayes' Rule. The bid is the *smallest* bid that can be given such that the probability that the bid is greater than the largest winning bid seen so far, conditioned on the opponent's bidding behaviour and the winners of previous rounds, is greater than some threshold value α. In this way, the bid utilises the distribution of previous bidding behaviours of the adversaries, and the past winning bids that the agent has seen so far, to attempt to predict the adversary's next bids and outbid them. This is explained in **Section II-B**.

After this bid is computed, the standard deviation of the normal distribution used by the Bayesian model is then updated to reflect the dynamic bidding landscape of the auction, allowing the agent to better react and adapt to its adversaries' bidding behaviours. This is explained further in **Section III**.

Finally, before the computed bid is returned to the Auctioneer, it passes through a series of heuristic rules, formulated using the knowledge base of the auction environment. These rules constrain the bid to account for patience, budget management, and future bidding opportunities. This is explained in **Section II-A**.

A. Knowledge Base Representation

Let $\mathcal{P}$ be the 'painter space', i.e. the set of all painters. Then $P, G, R, V \in \mathcal{P}$ represent the painters *Picasso*, *Van Gogh*, *Rembrandt*, and *Da Vinci*, respectively. Further define $\mathcal{B}$ as the agent space.

Then

$$\text{paintings} : \mathcal{B} \times \mathcal{P} \to \mathbb{N}_0$$

Identifies how many paintings of a particular painter $x \in \mathcal{P}$ that agent $b \in \mathcal{B}$ owns,

$$m : \mathcal{B} \to \mathbb{Z}$$

Identifies how much money an agent $b \in \mathcal{B}$ has,

$$\text{sale} : \mathbb{N}_0 \to \mathcal{P}$$

Identifies what painting $k \in \mathcal{K}$ (painting space) is for sale in round $r \in \mathbb{N}_0$, and finally

$$\text{bid} : \mathcal{B} \times \mathbb{N}_0 \to \mathbb{Z}$$

Is the bid function for agent $b \in \mathcal{B}$ for round $r \in \mathbb{N}_0$. Following this, universal rules that apply to all agents in the auction may be applied:

Rule 1. Every round, the highest bid wins.

$$\forall r \in R \bullet \left(\text{winner}(r) = \arg\max_{b \in \mathcal{B}} \big(\text{bid}(b, r)\big) \right)$$

Rule 2. For every agent, their bid must always be less than or equal to their budget.

$$\forall b \in \mathcal{B} \bullet \big(\forall r \in R \bullet (\text{bid}(b, r) \leq \text{budget}(b)) \big)$$

Then, finally, rules may be formalised that are specific to the strategic agent being designed, $s \in \mathcal{B}$. A strategically reasonable agent should adhere to the following rules.

R_1 **- Rule 1.** If $\text{paintings}(s, x) = 0$ for any P, G, R, V, then s should always place a bid as owning no paintings of a painter results in 0 probability of winning.

$$\forall r \in R \bullet \big(\text{sale}(r) \in \{x \in \{P, G, R, V\} \mid \text{paintings}(s, x) = 0\} \implies \text{bid}(s, r) > 0 \big)$$

R_2 **- Rule 2.** In each round, the upper bound of s' bid should be constrained as follows

$$\forall r \in R \bullet \left(\text{bid}(s, r) \leq m(s) \cdot \frac{1}{\text{paintings}(s, \text{sale}(r)) + 1} \right)$$

R_3 **- Rule 3.** Finally, a 'pacing' rule is defined in which the bid is bound by certain factors such as score size, number of opportunities to buy a certain painting in the future, and the number of painters of whom the agent owns zero paintings.

$$\beta \cdot \frac{m(s)}{\max(1, N_p) \cdot \max(1, \text{paintings}(s, \text{sale}(r)))},$$

Where N_p is the number of future rounds that will also sell painting $\text{sale}(r)$, and β is a scaling factor. Furthermore, because **Rules 2 and 3** are both upper-bound constraints, only the tighter of the two, per round, is applied: $\min(R_2, R_3)$

B. Bayesian Inference

Let $w_r = (m \in \mathbb{Z})_{i=1}^{r}$ be a sequence of winning bids for each round, up to some round r. Let B be the bid made by $s \in \mathcal{B}$ in round r. Then, finally, let θ be the opponents' bidding behaviours. Then

$$\mathbf{P}(\theta | w_{r-1}) = \frac{\mathbf{P}(w_{r-1} | \theta) \cdot \mathbf{P}(\theta | w_{r-2})}{\mathbf{P}(w_{r-1})}$$

The behaviour of opponents for round r is based on the winning bids sequence from rounds $[1, r-1]$.

Then, a decision rule may be applied to determine B with some threshold value α:

$$B = \min \left(\left\{ B' \in \mathbb{Z} \mid \mathbf{P}\big(B' > \max(w_r) \mid \theta, w_{r-1}\big) \geq \alpha \right\} \right)$$

This determines the minimum $\text{bid}(s, r)$ that s must make such that the probability that the bid is greater than that of its opponents, based on the previous winning bids, is greater than some threshold value α (interpreted as the 'confidence'). In the implementation, $\alpha = 0.6$.

However, even then there is one last challenge remaining: What should the standard deviation σ be? The chosen value of σ is, in fact, highly important in this issue as it determines the range of final bids available to the agent. For example, with $\sigma = 10$ and a single opponent whose posterior mean has converged to $\hat{\mu} \approx 10$, the resulting bid is approximately 12–13.

A fixed-σ approach was initially used for simplicity, however it was realised that this resulted in an agent that could not react properly and flexibly to any adversary it encountered. Between a flat bot 10's bid of 10, and another adversary's bid of between 20 and 40, one single, fixed σ cannot account for all of these possible bids. Thus, an adaptive-σ approach was implemented.

III. Adaptive-σ

The adaptive-σ approach was relatively intuitive to formulate and develop. It must allow σ to change such that the agent's bid can adapt to its adversaries' 'best' bids. So, by observing the past x winning bids (where $0 \leq x \leq r$ and r is the current round), σ for round $r+1$ is nothing but the standard deviation of the past x winning bids. In this way, the agent can observe the bidding behaviour of each previous winner and adapt its bids accordingly. If a flat-agent has consistently been winning, then σ will slowly be pushed down to respond to this, allowing the agent to reach competitive bids very quickly. Formally, taking the last x winning bids: $h_x = (h_{r-x}, h_{r-x+1}, \ldots, h_r)$. Then, $\sigma_{r+1} = \text{std}(h_x)$.

IV. Q-Learning & Lack of Effectiveness

Finally, Q-Learning was also explored during this assignment. It was implemented in the following way. The action of the agent was modelled as a specific scaling factor that can be applied to the bid. The Q-Learner chose an action $a \in \{0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.75\}$ and then the final, computed bid was $a \cdot \text{bid(s, r)}$. Further, every state $S \in \mathcal{S}$ was a tuple:

$$S = (C, m_{\text{bin}})$$

where $C \in \{0, 2, 5, 10, 25, 40, 41\}$ is the collection bin. Essentially, the score (using the multiplicative score function provided), was sorted into a specific category in C depending on its size. Finally:

$$m_{\text{bin}} = \begin{cases} 0, m(s) \leq \frac{1}{4}T \\ 1, \frac{1}{4}T < m(s) < \frac{3}{4}T \\ 2, m(s) \geq \frac{3}{4}T \end{cases}$$

Is a discrete natural number that indicates the 'level' of the budget (with $T \in \mathbb{Z}$ being the total original budget). Bins were utilised in order to decrease the size of the state-space and the subsequent Q-table.

However, Q-Learning was finally discarded as a component of the agent's bidding strategy as it did not result in auction-winning bids. For the majority of the time, even after extensive training and hyper-parameter tuning, the agent would either bid far too aggressively and discard its budget by round 100, or

it would bid too conservatively and not win enough paintings. It was eventually found that varying the Bayesian model's σ parameter was far more effective than attempting to apply a post-Bayesian scaling factor to the final bid. A fixed σ only provides competitive bids for certain environments and adversaries, so the agent had to rely on Q-Learner to either lower the Bayesian bid or artificially inflate it. But, because the table does not have enough actions to provide the high degree of flexibility required, the Q-Learner was rendered unable to scale the bid correctly. As such, it was determined that a far more intuitive and efficient solution was to create an adaptive algorithm to fine-tune σ during the auction.

V. Results

A. Tests

Testing the agent included creating a number of rooms with a variety of strategic adversaries. An initial baseline test was carried out with the agent competing only against the random agent. This serves as a useful benchmarking test as it can be used to observe whether the agent is 'better' than simply taking random bids. Seven other tests were then carried out, each with ten 200-round auctions, and performance was evaluated by observing the win-rate over all auctions. Each test involved generating a random assortment of agents (from a pre-determined list) to compete against the designed one, and the cumulative win count of the agent for each auction was recorded. The random assortment included agents with strategies that were specifically designed to exploit vulnerabilities in this 'adaptive-σ' approach, called 'purpose-made' adversaries. Such strategies included intentionally 'injecting' high bids in the early game, causing the designed agent's adaptive-σ bid to be inflated, thereby forcing the agent to overpay in subsequent rounds due to its application of a historical window. It should be noted that the only aim of such bids was to disrupt the agent's adaptive-σ, it was not a strategic bid in and of itself.

The win-rate for the agent in each test is provided in **Table I**. As stated above, the benchmarking test against random was carried out. Against a random bidder, the designed agent obtained a win-rate of 100 %. It can further be seen that the agent achieved a 100 % win-rate for three of seven tests. Importantly, for two of three tests, the agent obtained a 100 % win-rate when faced against two purpose-made adversaries. On the other test, T5, the designed agent obtained a win-rate of 20 %. Three of the tests (T2, T3, and T5) included the agent itself as an adversary. The agent obtained a 70 % win-rate in T2, 30 % in T3, and 10 % in T7.

Test Number	Composition of agents	Win Rate (%)
1	3 Adversaries, including two purpose-designed Agents	100
2	4 Adversaries, including itself and random	70
3	4 Adversaries, including itself and two purpose-designed Agents	30
4	2 Adversaries, including random	100
5	5 Adversaries, including itself and two purpose-designed Agents	20
6	3 Adversaries, including random	100
7	7 Adversaries, including itself, random, and two purpose-designed Agents	10

TABLE I
TABLE SHOWING WIN-RATE OF THE AGENT IN EACH TEST.

B. Analysis of Results

While the agent demonstrated a 100 % win rate against randomized opponents, empirical analysis of the more diverse tests (such as Test 3, wherein the agent achieved a 30 % win rate) revealed a consistent, multi-factor cause-of-loss in the bidding strategy. These factors were: Bayesian variance interference or 'injection', pacing constraint over-correction, and the multiplicative scoring penalty.

Firstly, the adaptive-σ algorithm proved vulnerable to certain exploitative bidding strategies. The purpose-built agents were designed to place larger bids in the opening 15 rounds, rapidly exhausting their initial 1001 budget. Due to the size of the historical window used for σ computation, long after the aggressive agents were eliminated from contention (due to lack of budget), the agent's Bayesian model continued to predict inflated bid thresholds, causing it to regularly overpay against remaining opponents who were bidding far less.

Secondly, this early-game budget exhaustion triggered a severe over-correction in the agent's heuristic pacing constraints. As defined in **Rule 2**, the agent's maximum allowable bid is strictly bounded by a fraction of its remaining budget. Having depleted nearly half of its budget in the first 40 rounds, the agent was rendered unable to justify even medium-sized bids for the remaining 160 rounds. The upper bound was forced to be reduced to < 10, resulting in the agent having to rely only on winning largely uncontested paintings near the end for cheap bids.

Finally, this behaviour actively undermined the auction's core winning condition. The auction's scoring function rewards more uniform distribution more than unbalanced collections. By winning cheap paintings late in the auction, the agent finished with highly skewed collections. In contrast, other adversaries maintained stable bids throughout the entire auction, achieving more balanced portfolios. Even when the agent won a greater absolute number of paintings, the score of its unbalanced set resulted in its inability to win.

In future iterations, the Bayesian model should integrate opponent budget tracking. By capping the computed bid at $\max(\text{opponent_budgets})+1$, the agent can secure a win without being overly influenced by historical variance 'injection'. Furthermore, a dynamic window size would allow the agent to adapt properly to significant variance spikes and stabilise σ much faster. Another method to mitigate these exploitations would be to remove sudden variance spikes, resulting in a more robust σ algorithm.